

Matrices: Engineering Mathematics Study Notes

2. Gauss-Jordan Method

High-Level Overview

The **Gauss-Jordan Method** is a direct extension and modification of the standard Gauss-Elimination method. While Gauss-Elimination stops once the matrix is transformed into an upper-triangular form (requiring back-substitution), the Gauss-Jordan method uses elementary row operations to reduce the augmented matrix all the way into a **diagonal matrix** (specifically, an identity matrix on the left side).

In this diagonalized form, each equation directly involves only a single unknown, meaning the values for variables like x, y, z (and w) can be read straight off the final augmented column without needing any back-substitution steps.

Step-by-Step Procedure

1. **Form the Augmented Matrix:** Write the given system of linear equations in matrix form $AX = B$, and combine them into the augmented matrix $[A \mid B]$.
 2. **Forward Elimination (Making Zeros Below the Diagonal):** Use elementary row operations to create zeros below the main diagonal, similar to the initial steps of Gauss elimination.
 3. **Diagonalization (Making Zeros Above the Diagonal):** Continue applying row operations to eliminate non-zero elements *above* the main diagonal, ensuring the coefficient matrix on the left transforms into an identity matrix (I).
 4. **Read the Solution:** Once the left side is reduced to the identity matrix, the right-hand column directly gives the values of the unknowns.
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Worked Examples

Example 1: Solving a 3×3 System

Solve the system of equations using the Gauss-Jordan method:

$$x + y + z = 9$$

$$2x - 3y + 4z = 13$$

$$3x + 4y + 5z = 40$$

Step 1: Write the Augmented Matrix

$$[A \mid B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 9 \\ 2 & -3 & 4 & 13 \\ 3 & 4 & 5 & 40 \end{array} \right]$$

Step 2: Apply Row Operations to Create Zeros in the First Column

Perform $R_2 \rightarrow R_2 - 2R_1$ and $R_3 \rightarrow R_3 - 3R_1$:

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 9 \\ 0 & -5 & 2 & -5 \\ 0 & 1 & 2 & 13 \end{array} \right]$$

Step 3: Normalize the Pivot in Row 2 and Eliminate

Perform $R_2 \rightarrow -\frac{1}{5}R_2$:

$$\sim \begin{bmatrix} 1 & 1 & 1 & : & 9 \\ 0 & 1 & -2/5 & : & 1 \\ 0 & 1 & 2 & : & 13 \end{bmatrix}$$

Perform $R_3 \rightarrow R_3 - R_2$:

$$\sim \begin{bmatrix} 1 & 1 & 1 & : & 9 \\ 0 & 1 & -2/5 & : & 1 \\ 0 & 0 & 12/5 & : & 12 \end{bmatrix}$$

Step 4: Scale Row 3 to Get a Leading 1

Perform $R_3 \rightarrow \frac{5}{12}R_3$:

$$\sim \begin{bmatrix} 1 & 1 & 1 & : & 9 \\ 0 & 1 & -2/5 & : & 1 \\ 0 & 0 & 1 & : & 5 \end{bmatrix}$$

Step 5: Back-Elimination (Creating Zeros Above the Diagonal)

Eliminate the entry above in row 2 using $R_2 \rightarrow R_2 + \frac{2}{5}R_3$:

$$\sim \begin{bmatrix} 1 & 1 & 1 & : & 9 \\ 0 & 1 & 0 & : & 3 \\ 0 & 0 & 1 & : & 5 \end{bmatrix}$$

Eliminate entries in row 1 using $R_1 \rightarrow R_1 - R_2$:

$$\sim \begin{bmatrix} 1 & 0 & 1 & : & 6 \\ 0 & 1 & 0 & : & 3 \\ 0 & 0 & 1 & : & 5 \end{bmatrix}$$

Finally, eliminate the remaining non-zero entry in row 1 using $R_1 \rightarrow R_1 - R_3$:

$$\sim \begin{bmatrix} 1 & 0 & 0 & : & 1 \\ 0 & 1 & 0 & : & 3 \\ 0 & 0 & 1 & : & 5 \end{bmatrix}$$

Final Answer:

$$x = 1, \quad y = 3, \quad z = 5$$

Example 2: Another 3×3 System

Solve the system of equations using the Gauss-Jordan method:

$$x + y + z = 3$$

$$2x + 3y + z = 6$$

$$x - y - z = -3$$

Step 1: Write the Augmented Matrix

$$[A \mid B] = \begin{bmatrix} 1 & 1 & 1 & : & 3 \\ 2 & 3 & 1 & : & 6 \\ 1 & -1 & -1 & : & -3 \end{bmatrix}$$

Step 2: Eliminate Below the First Pivot

Perform $R_2 \rightarrow R_2 - 2R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$\sim \begin{bmatrix} 1 & 1 & 1 & : & 3 \\ 0 & 1 & -1 & : & 0 \\ 0 & -2 & -2 & : & -6 \end{bmatrix}$$

Step 3: Continue Row Operations

Perform $R_1 \rightarrow R_1 - R_2$ and $R_3 \rightarrow R_3 + 2R_2$:

$$\sim \begin{bmatrix} 1 & 0 & 2 & : & 3 \\ 0 & 1 & -1 & : & 0 \\ 0 & 0 & -4 & : & -6 \end{bmatrix}$$

Step 4: Scale Row 3

Perform $R_3 \rightarrow -\frac{1}{4}R_3$:

$$\sim \begin{bmatrix} 1 & 0 & 2 & : & 3 \\ 0 & 1 & -1 & : & 0 \\ 0 & 0 & 1 & : & 3/2 \end{bmatrix}$$

Step 5: Complete Diagonalization

Perform $R_1 \rightarrow R_1 - 2R_3$ and $R_2 \rightarrow R_2 + R_3$:

$$\sim \begin{bmatrix} 1 & 0 & 0 & : & 0 \\ 0 & 1 & 0 & : & 3/2 \\ 0 & 0 & 1 & : & 3/2 \end{bmatrix}$$

Final Answer:

$$x = 0, \quad y = \frac{3}{2}, \quad z = \frac{3}{2}$$

Example 3: A 4×4 System

Solve the system of equations using the Gauss-Jordan method:

$$2x - 2y = 1$$

$$-x + 2y - 3z = -2$$

$$-2y + 2z - 4w = -1$$

$$z - w = -3$$

Step 1: Write the Augmented Matrix

$$[A \mid B] = \begin{bmatrix} 2 & -2 & 0 & 0 & : & 1 \\ -1 & 2 & -3 & 0 & : & -2 \\ 0 & -2 & 2 & -4 & : & -1 \\ 0 & 0 & 1 & -1 & : & -3 \end{bmatrix}$$

Step 2: Normalize Row 1 PivotPerform $R_1 \rightarrow \frac{1}{2}R_1$:

$$\sim \begin{bmatrix} 1 & -1 & 0 & 0 & : & 1/2 \\ -1 & 2 & -3 & 0 & : & -2 \\ 0 & -2 & 2 & -4 & : & -1 \\ 0 & 0 & 1 & -1 & : & -3 \end{bmatrix}$$

Step 3: Systematic Elimination across RowsPerform $R_2 \rightarrow R_2 + R_1$:

$$\sim \begin{bmatrix} 1 & -1 & 0 & 0 & : & 1/2 \\ 0 & 1 & -3 & 0 & : & -3/2 \\ 0 & -2 & 2 & -4 & : & -1 \\ 0 & 0 & 1 & -1 & : & -3 \end{bmatrix}$$

Perform $R_3 \rightarrow R_3 + 2R_2$:

$$\sim \begin{bmatrix} 1 & -1 & 0 & 0 & : & 1/2 \\ 0 & 1 & -3 & 0 & : & -3/2 \\ 0 & 0 & -4 & -4 & : & -4 \\ 0 & 0 & 1 & -1 & : & -3 \end{bmatrix}$$

Scale Row 3 using $R_3 \rightarrow -\frac{1}{4}R_3$:

$$\sim \begin{bmatrix} 1 & -1 & 0 & 0 & : & 1/2 \\ 0 & 1 & -3 & 0 & : & -3/2 \\ 0 & 0 & 1 & 1 & : & 1 \\ 0 & 0 & 1 & -1 & : & -3 \end{bmatrix}$$

Perform $R_4 \rightarrow R_4 - R_3$:

$$\sim \begin{bmatrix} 1 & -1 & 0 & 0 & : & 1/2 \\ 0 & 1 & -3 & 0 & : & -3/2 \\ 0 & 0 & 1 & 1 & : & 1 \\ 0 & 0 & 0 & -2 & : & -4 \end{bmatrix}$$

Scale Row 4 using $R_4 \rightarrow -\frac{1}{2}R_4$:

$$\sim \begin{bmatrix} 1 & -1 & 0 & 0 & : & 1/2 \\ 0 & 1 & -3 & 0 & : & -3/2 \\ 0 & 0 & 1 & 1 & : & 1 \\ 0 & 0 & 0 & 1 & : & 2 \end{bmatrix}$$

Step 4: Backward Elimination to Form Identity MatrixPerform $R_3 \rightarrow R_3 - R_4$:

$$\sim \begin{bmatrix} 1 & -1 & 0 & 0 & : & 1/2 \\ 0 & 1 & -3 & 0 & : & -3/2 \\ 0 & 0 & 1 & 0 & : & -1 \\ 0 & 0 & 0 & 1 & : & 2 \end{bmatrix}$$

Perform $R_2 \rightarrow R_2 + 3R_3$:

$$\sim \begin{bmatrix} 1 & -1 & 0 & 0 & : & 1/2 \\ 0 & 1 & 0 & 0 & : & -9/2 \\ 0 & 0 & 1 & 0 & : & -1 \\ 0 & 0 & 0 & 1 & : & 2 \end{bmatrix}$$

Perform $R_1 \rightarrow R_1 + R_2$:

$$\sim \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & -4 \\ 0 & 1 & 0 & 0 & -9/2 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 2 \end{array} \right]$$

Final Answer:

$$x = -4, \quad y = -\frac{9}{2}, \quad z = -1, \quad w = 2$$